Introduction

The World Is a Chessboard

A Strategic Reality

We often think life is a series of random events. **It is not.** It is a game, sometimes cooperative, sometimes competitive, but *always strategic*.

- From the vegetable vendor negotiating prices ...
- ... to the diplomat averting war.
- **We are all players.**
- The question is: are you playing with a strategy?

Who This Is For

- Professionals & Managers
- Negotiators & Leaders
- Entrepreneurs & Investors
- Curious Minds
- *Anyone who interacts with other humans*

What Is Game Theory?

Definition

Game theory is the study of **strategic interdependence**: situations where your outcome depends not only on what you do, but on what others choose to do.

- **Players**: Individuals, firms, nations, algorithms
- **Strategies**: Plans of action, sometimes contingent
- **Payoffs**: Outcomes that players value, qualitative or quantitative (money, safety, reputation)
- **Information**: What players know (rarely perfect)
- **Equilibrium**: Stable state where strategies are mutual best responses

Core Insight

Success depends on anticipating others' choices and, when possible, shaping them.

- **Not** about controlling nature or solving physics
- **About** responding to decision-makers who respond
- Applies wherever outcomes are shaped by choices
- From salary negotiations to nuclear deterrence

A Brief History

Origins

- **1944**, Von Neumann & Morgenstern publish *Theory of Games and Economic Behaviour*
- **1950**, John Nash introduces the *Nash Equilibrium*
- **1994**, Nash, Harsanyi & Selten win the Nobel Prize in Economics
- **2005, 2012**, Further Nobel awards for Mechanism Design and Matching Theory

Key Pioneers

- **John von Neumann**, Founder of game theory
- **John Nash**, Equilibrium concept
- **Thomas Schelling**, Focal points & brinkmanship
- **Kahneman & Tversky**, Behavioural economics
- **Kautilya (Chanakya)**, Ancient strategic wisdom

Why Game Theory Matters Today

Traditional Thinking

- Assumes only you are in control
- Optimises in isolation
- "How do I get the best outcome?"
- *Like an engineer solving physics*

Game-Theoretic Thinking

- Assumes others are strategic too
- Optimises in interaction
- "How do I get the best outcome **given what others will do?**"
- *Like a chess player anticipating moves*

Why Game Theory Matters Today

Where It Applies

- **Business**, Pricing, competition, M&A
- **Politics**, Elections, coalitions, treaties
- **Personal Life**, Negotiation, relationships
- **Technology**, Platform strategy, auctions
- **Society**, Public goods, voting, norms
- **Finance**, Auctions, trading, insurance

Key Insight

You are **never** playing solitaire (i.e. alone). You are always playing chess (i.e. against someone).

How to Use This Playbook

Structure of Each Concept

- **Introduction**, What is it?
- **Situation**, When does it apply?
- **Steps**, How to use it?
- **Examples**, Real-world anchors
- **Takeaway**, The bottom line

Sections Covered

- Foundational Concepts
- Game Types & Structures
- Decision-Making Approaches
- Credibility & Commitment
- Strategic Interaction in Relationships
- Competitive Advantage
- Information & Uncertainty
- Negotiation & Bargaining
- Market Design
- Collective Challenges
- Prospect Theory

How to Use This Playbook

A Word of Warning

Game Theory is a **map**, not the territory. Real life is messy. People are irrational. Rules change. Use these models as lenses, not as algorithms.

The Infinite Game

In the "Finite Game" of a football match, you play to *win*. In the "Infinite Game" of life, you play to *keep playing*. Understand the incentives. Respect the players.

Foundational Concepts

Understanding Payoff Matrices

What is a Payoff Matrix?

- A table showing outcomes for all strategy combinations
- Rows represent Player 1's choices (Ramesh)
- Columns represent Player 2's choices (Suresh)
- Each cell shows the outcome when both make choices

Example: Coffee Pricing

Ramesh \ Suresh	High Price	Low Price
High Price	(50, 50)	(20, 70)
Low Price	(70, 20)	(30, 30)

Payoffs in Rs 1000s of profit **How to Read the Matrix:**

- First number = Ramesh's profit, second = Suresh's profit
- **Example 1:** If both choose **High Price**, we look at top-left cell:
  $$(50, 50)$$
  Both earn Rs 50,000 each
- **Example 2:** If Ramesh chooses **Low Price** and Suresh chooses **High Price**, we look at:
  $$(70, 20)$$
  Ramesh earns Rs 70,000, Suresh earns Rs 20,000
- This shows how one player's decision affects the other

Finding Dominant Strategies

What is a Dominant Strategy?

- A strategy that gives you a better payoff *for every possible move* of your opponent
- It works **no matter what the opponent chooses**
- So you don't need to guess or predict anything

Example Matrix:

You \ Opponent	A	B
Option 1	(3, 3)	(1, 4)
Option 2	**(4, 2)**	**(2, 3)**

How to Think About It:

- Ask: "If my opponent picks A or B, which option gives me more?"
- **Case 1: Opponent chooses A**
  - Option 1 gives you 3
  - Option 2 gives you 4
  - ⇒ Option 2 is better
- **Case 2: Opponent chooses B**
  - Option 1 gives you 1
  - Option 2 gives you 2
  - ⇒ Option 2 is again better
- **Key Insight:** No matter what opponent does (A or B), Option 2 always gives you a higher payoff
- **Conclusion:** Option 2 is a dominant strategy

Finding Nash Equilibrium: Idea

What is Nash Equilibrium?

- Each player is choosing their **best response** to the other
- No one wants to change their decision *alone*
- Think: "Am I doing the best given what the other person is doing?"

Example Matrix:

Player 1 \ Player 2	Left	Right
Up	(5, 5)	(1, 4)
Down	(4, 1)	(2, 2)

First number = Player 1, second = Player 2

Step 1: Find Best Responses

For Player 1 (compare numbers in each column):

- If Player 2 plays Left: compare 5 vs 4 ⇒ <u>5 is higher</u>
- If Player 2 plays Right: compare 1 vs 2 ⇒ <u>2 is higher</u>

For Player 2 (compare numbers in each row):

- If Player 1 plays Up: compare 5 vs 4 ⇒ <u>5 is higher</u>
- If Player 1 plays Down: compare 1 vs 2 ⇒ <u>2 is higher</u>

Marking in the matrix:

	Left	Right
Up	($\underline{5}$, $\underline{5}$)	(1, 4)
Down	(4, 1)	($\underline{2}$, $\underline{2}$)

Step 2: Find Nash Equilibrium

Key Idea:

- Nash equilibrium occurs where **both players are choosing best responses**
- That means: both numbers in the cell are "selected" (underlined)

	Left	Right
Up	($\underline{5}$, $\underline{5}$)	(1, 4)
Down	(4, 1)	($\underline{2}$, $\underline{2}$)

Observation:

- Two cells have both values underlined:
  - (Up, Left) ⇒ (5,5)
  - (Down, Right) ⇒ (2,2)

Conclusion:

- This game has **two Nash equilibria**

Is (5,5) a Nash Equilibrium?

Short Answer: Yes, it *is* a Nash Equilibrium

Why?

- At (Up, Left):
  - Player 1 gets 5 — switching to Down gives 4 (worse)
  - Player 2 gets 5 — switching to Right gives 4 (worse)
- So neither player wants to deviate

Then why focus on (2,2)?

- (2,2) is also a Nash equilibrium
- But it is **worse for both players**
- This shows:
  - Nash equilibrium is about **stability**, not **optimality**

Key Insight:

- Players may get stuck in a worse outcome even when a better one exists

Which Nash Equilibrium Will Be Chosen?

Recall: We found two Nash equilibria

	Left	Right
Up	(5, 5)	(1, 4)
Down	(4, 1)	(2, 2)

Question: Which outcome will players choose?

- Both (5,5) and (2,2) are stable (no one wants to deviate alone)
- **(5,5) is better for both**
  - Higher payoff for Player 1 and Player 2
- But there is a problem: **coordination**
  - To reach (5,5), both must choose (Up, Left)
  - If one player makes a mistake, payoff drops sharply
- **(2,2) is safer**
  - Even if unsure about the other player, risk is lower

Key Insight:

- Some equilibria are **better**, others are **safer**
- Real-world outcomes depend on **trust, expectations, and coordination**

Coordination Problem: Risk vs Reward

Think like a player:

- "I would like (5,5)... but what if the other player doesn't cooperate?"

Compare the risk:

- If you aim for (5,5) but the other player deviates:
  - You may end up with **1** (very low payoff)
- If you aim for (2,2) and the other deviates:
  - Your payoff does not fall as drastically

Interpretation:

- (5,5) = **High reward, high risk**
- (2,2) = **Lower reward, safer choice**

Real-world analogy:

- Two firms setting high prices (cooperate) vs undercutting each other
- Trust leads to better outcomes, but fear leads to safer ones

How to Reach the Better Equilibrium (5,5)?

Problem:

- (5,5) is better for both players
- But it requires **coordination and trust**

What can help players coordinate?

- **Communication**
  - Players can agree beforehand: "Let's both choose (Up, Left)"
- **Repeated Interaction**
  - If the game is played many times, cooperation can be rewarded
  - Deviating once may lead to punishment later
- **Trust / Reputation**
  - If a player is known to cooperate, others are more likely to do the same
- **Focal Points (Obvious Choices)**
  - Players may naturally gravitate to "prominent" outcomes like (5,5)

Key Idea:

- Better outcomes often require **coordination mechanisms**, not just rational thinking

What If Coordination Fails?

Without coordination:

- Players may avoid risk and choose safer strategies
- This can lead to the equilibrium:

$$(2, 2)$$

Why does this happen?

- Fear of mismatch:
  - "What if I choose Up but the other chooses Right?"
- Lack of trust:
  - No guarantee the other player will cooperate
- No communication:
  - Players cannot coordinate their choices

Final Insight:

- Rational players can end up in **worse outcomes**
- The issue is not intelligence, but **lack of coordination**

Real-World Example: Traffic Coordination

Imagine: Choosing which side of the road to drive on

Driver 1 \ Driver 2	**Left**	**Right**
Left	(5, 5)	(0, 0)
Right	(0, 0)	(5, 5)

- If both choose the **same side**, traffic flows smoothly → high payoff (5,5)
- If they choose **different sides**, there is chaos (or accidents!) → payoff (0,0)

Key Observations:

- There are **two Nash equilibria**:
  - (Left, Left)
  - (Right, Right)
- Both are equally good—but players must **coordinate**

Why Do We Successfully Coordinate?

Question: Why don't drivers crash every day?

- **Social conventions**
  - In India, everyone follows "drive on the left"
- **Clear rules and laws**
  - Government enforces one equilibrium
- **Common expectations**
  - You expect others to follow the same rule

Connection to Game Theory:

- Real-world systems often **select one equilibrium**
- Once established, everyone follows it automatically

Final Insight:

- Good outcomes are not just about incentives
- They depend on **rules, norms, and coordination mechanisms**

Popular Strategies

Nash Equilibrium: The Stable State

- **Introduction**: A state where no player can improve their outcome by changing their strategy alone. It is the point at which the game *self-locks*, often into a suboptimal outcome for all.
- **Situation**: Arises in any repeated interaction where actors respond to each other, arms races, price wars, social norms, congestion. Whenever you are stuck in a bad pattern everyone perpetuates.
- **Steps**:
  - List all players and their possible strategies
  - Map payoffs for each combination
  - For each player, find the best response to every opponent strategy
  - Identify where best responses align, that is the equilibrium
  - Ask: is it efficient? If not, what changes the game?
- **Examples**: Everyone stands at a cricket match so no one sees better; arms races leaving all nations poorer; airlines locked in price wars destroying profitability.

Nash Equilibrium: Ramesh-Suresh Coffee Shop Example

The Scenario:

- Two coffee shops across the street from each other
- Originally both charged Rs 100 per cup, profits were stable
- Each can independently choose to charge Rs 100 or Rs 70
- If one cuts price while other maintains, the cutter gains customers
- Both are stuck at Rs 70 now : neither can improve alone

Payoff Matrix (Daily Profit):

Ramesh \ Suresh	**Rs 100**	**Rs 70**
Rs 100	(5k, 5k)	(1k, 7k)
Rs 70	(7k, 1k)	**(3k, 3k)**

Analysis Steps:

- **Check Best Responses:**
  - If Suresh charges Rs 100, Ramesh's best = Rs 70 (7k > 5k)
  - If Suresh charges Rs 70, Ramesh's best = Rs 70 (3k > 1k)
  - Same logic for Suresh
- **Nash Equilibrium:** Both charge Rs 70 (bold in matrix)
- **The Trap:** Each earns only Rs 3000 instead of Rs 5000
- **Key Insight:** No player can improve by changing alone : the equilibrium is stable but suboptimal

Dominant Strategy: The Safe Move

- **Introduction**: A dominant strategy delivers the highest payoff for a player *regardless of what others do*. It is the "no-brainer", you need not read the opponent's mind when you have one.
- **Situation**: Look for dominant strategies before trying to anticipate opponents. Useful in competitive exams, auctions, market pricing, and any situation where one choice outperforms all others unconditionally.
- **Steps**:
  - Draw a payoff matrix
  - For each of your strategies, find the payoffs against every opponent move
  - Check if one row always beats the others
  - If yes, that is your dominant strategy, play it
  - If not, identify your opponent's dominant strategy and best-respond to it
- **Examples**: Amazon's relentless cost leadership; index fund investing over stock-picking; vaccination during a pandemic regardless of what neighbours do.

Dominant Strategy: Ramesh-Suresh Wi-Fi Example

The Scenario:

- Ramesh considers adding high-speed Wi-Fi to his coffee shop
- If Suresh doesn't install Wi-Fi, Ramesh attracts students & freelancers
- If Suresh does install Wi-Fi, Ramesh needs it to stay competitive
- Either way, installing Wi-Fi makes Ramesh better off
- This is a dominant strategy : works regardless of opponent's choice

Payoff Matrix (Weekly Additions):

Ramesh \ Suresh	**No Wi-Fi**	**Install Wi-Fi**
No Wi-Fi	(50, 50)	(20, 90)
Install Wi-Fi	**(100, 30)**	**(60, 60)**

Analysis Steps:

- **Test Ramesh's Options:**
  - If Suresh has no Wi-Fi: Install (100) > Don't (50)
  - If Suresh has Wi-Fi: Install (60) > Don't (20)
- **Dominant Strategy:** Install Wi-Fi (bold row) : always better regardless of Suresh's choice
- **Same Logic for Suresh:** Installing Wi-Fi dominates for him too
- **Key Insight:** No mind-reading required. The strategy is robust : it's insurance, not cleverness

Zero-Sum Games: Winner-Takes-All

- **Introduction**: In a zero-sum game the total available benefit is fixed, one player's gain is another's exact loss. These are games of pure conflict where cooperation is mathematically impossible.
- **Situation**: Applicable where resources are genuinely scarce and non-expandable: elections, IIT seats, market share in a stagnant industry, athletic competitions, inheritance division.
- **Steps**:
  - Verify the game is truly zero-sum before treating it as one
  - Identify your relative strengths vs. opponents
  - Use minimax strategy, minimise maximum possible loss
  - Look for dominant positions to exploit
  - Consider whether the game can be redesigned as positive-sum
- **Examples**: Poker tables; election seat counts; dividing a fixed estate; ranking systems where one rise means another's fall.

Zero-Sum Games: Ramesh-Suresh Customer War

The Scenario:

- Both locked in war over same 50 factory workers who cross street every morning
- Fixed customer pool, when one gains a customer, other loses one
- Discounts, louder music, bigger signboards
- Every move designed to steal, not to create
- Victory for one = defeat for other
- Classic zero-sum: customer pool is fixed

Payoff Matrix (Daily Customers):

Ramesh \ Suresh	Normal	Aggressive
Normal	(25, 25)	(15, 35)
Aggressive	(35, 15)	(25, 25)

Analysis Steps:

- **Check Total:** Each cell sums to 50 customers (25+25, 15+35, etc.)
- **Zero-Sum Property:** One's gain = Other's exact loss
- **No Win-Win:** Cannot both gain customers simultaneously
- **Pure Conflict:** Cooperation doesn't expand the pie
- **Key Insight:** Verify game is truly zero-sum before treating it as one, many situations that seem zero-sum are actually positive-sum

Positive-Sum Games: Creating Mutual Benefit

- **Introduction**: Positive-sum games allow the total gains to grow, "growing the pie" rather than fighting over slices. Trade, collaboration, and innovation are classic examples where mutual gain is possible.
- **Situation**: Most business partnerships, trade agreements, marriages, and open-source projects. When both parties gain from interaction, the strategic goal shifts from defeating to creating and coordinating.
- **Steps**:
  - Map potential joint gains from cooperation
  - Identify each party's comparative advantage
  - Design a split of gains that both prefer over non-cooperation
  - Build trust mechanisms to prevent defection
  - Iterate, reputation and repeat interaction sustain positive-sum outcomes
- **Examples**: Bangalore's tech ecosystem where new firms benefit incumbents; open-source software; international trade agreements; ridesharing platforms benefiting both drivers and riders.

Positive-Sum: Ramesh-Suresh Grow Together

The Scenario:

- Both improve coffee, seating, vibe
- Street becomes destination
- Office workers detour on purpose, weekend crowds arrive
- Better beans from suppliers, local bakery partnerships
- Each one's improvements lift expectations for both
- Total customers increase instead of staying fixed

Payoff Matrix (Daily Customers):

Ramesh \ Suresh	Low Quality	High Quality
Low Quality	(25, 25)	(30, 40)
High Quality	(40, 30)	**(55, 55)**

Analysis Steps:

- **Check Totals:**
  - Both Low: 50 total
  - Mixed: 70 total
  - Both High: 110 total!
- **Positive-Sum Property:** Pie grows with cooperation
- **Best Outcome:** Both High Quality (55, 55)
- **Key Insight:** Real question isn't how to beat rival, it's how to grow the total market
- **Biggest wins:** Come from making location a destination

Prisoner's Dilemma: The Trust Problem

- **Introduction**: The most famous paradox in game theory, individual rationality (self-protection) leads to collective irrationality (both suffer). Mutual cooperation would be best, but each defects, fearing betrayal.
- **Situation**: Any situation where independent self-interest produces a worse collective outcome: price wars, arms races, doping in sports, environmental pollution, negative political campaigning.
- **Steps**:
  - Recognise the dilemma structure in your situation
  - Assess the shadow of the future, is there a next interaction?
  - Build communication and trust mechanisms
  - Introduce contracts or penalties for defection
  - Use tit-for-tat strategies in repeated play
- **Examples**: Two sweet shops undercutting each other to thin margins; competing nations in arms races; athletes using performance-enhancing drugs; public littering.

Prisoner's Dilemma: Ramesh-Suresh Price War

The Scenario:

- Both shops initially charge fair prices and maintain quality
- Each can choose: *Cooperate* (fair pricing) or *Defect* (cut prices aggressively)
- If both cooperate: stable profits, good margins
- If one defects while other cooperates: defector gains market share
- If both defect: price war destroys profitability for both
- Fear of betrayal drives both to defect

Payoff Matrix (Monthly Profit in Rs):

Ramesh \ Suresh	Cooperate	Defect
Cooperate	(50k, 5k0)	(20k, 70k)
Defect	(70k, 20k)	**(30k, 30k)**

Analysis Steps:

- **Check Incentives:**
  - If Suresh cooperates: Defect (70K) > Cooperate (50K)
  - If Suresh defects: Defect (30K) > Cooperate (20K)
- **Dominant Strategy:** Defect for both players
- **Nash Equilibrium:** (Defect, Defect) : bold in matrix
- **The Paradox:** Both earn 30K instead of 50K
- **Escape Routes:** Repeated interaction, trust-building, contracts, tit-for-tat

Chicken Game: The Dangerous Standoff

- **Introduction**: A test of nerves where the worst outcome is mutual destruction. Unlike the Prisoner's Dilemma, someone *must* yield. The winner is the one who convincingly cannot back down.
- **Situation**: High-stakes confrontations where one party must blink: geopolitical crises, union vs. management standoffs, legislative gridlock, corporate takeover battles, sovereign debt negotiations.
- **Steps**:
  - Assess the cost of mutual destruction to both sides
  - Signal commitment credibly, make retreat costly or visible
  - Look for face-saving offramps for the opponent
  - Avoid symmetrical escalation
  - Know your red lines before the confrontation begins
- **Examples**: Cuban Missile Crisis; union strikes vs. management lockouts; government debt-ceiling standoffs; two buses on a narrow road.

Stag Hunt: Coordination Through Trust

- **Introduction**: A coordination game with two equilibria, a superior cooperative outcome (the Stag) and an inferior but safe solo outcome (the Hare). The challenge is not conflict but *assurance*: knowing others will commit.
- **Situation**: Climate accords, new technology standards, currency adoption, startup equity decisions, team sports defence, any endeavour where joint success requires full participation.
- **Steps**:
  - Communicate the payoff difference between cooperation and going solo
  - Build pre-commitment mechanisms
  - Start with a small, credible coalition
  - Provide insurance against defection
  - Create repeated-game conditions so defectors can be punished
- **Examples**: Nations adopting climate accords; tech industry standardising on USB-C; startups choosing equity over salary; dam-building village cooperatives.

Chicken Game: Ramesh-Suresh 24/7 Standoff

The Scenario:

- Both decide to stay open 24/7 to outlast each other
- Recklessly spending on electricity and overnight staff
- Like two trucks heading toward each other
- Winner is whoever has bigger bank account to absorb losses
- But if neither blinks, both go bankrupt
- Worst outcome: mutual destruction

Payoff Matrix (Monthly Net Profit/Loss in Rs):

Ramesh \ Suresh	**Normal Hours**	**24/7**
Normal Hours	(40k, 40k)	(10k, 55k)
24/7	(55k, 10k)	(-20k, -20k)

Analysis Steps:

- **Check Outcomes:**
  - Both normal: Good for both (40K each)
  - One blinks: Blinker loses badly (10K vs 55K)
  - Neither blinks: Both lose (Rs 20K)
- **Two Nash Equilibria:** (Normal, 24/7) and (24/7, Normal)
- **Key Difference:** Someone MUST yield
- **Winning Strategy:** Convince opponent you'll never back down
- **Real Danger:** Both commit and crash

Battle of the Sexes: Preference Conflicts

- **Introduction**: Both players want to coordinate but disagree on *which* coordination point to choose. It is not pure conflict, both want togetherness, but power determines whose preference wins.
- **Situation**: Merger decisions (who becomes CEO?), standard-setting wars (Mac vs PC), coalition government formation, meeting location choices, holiday planning.
- **Steps**:
  - Confirm both parties genuinely prefer coordination to no deal
  - Identify the current default, who benefits from the status quo?
  - Use side-payments or alternating concessions ("I'll watch cricket now, you choose next time")
  - Make concessions visible and reciprocal
  - Formalise agreements to avoid renegotiation
- **Examples**: Merger CEO selection; choosing a vacation destination; technology standard wars; political coalition ministry allocation.

Stag Hunt: Ramesh-Suresh Quality Upgrade

The Scenario:

- Both invest in quality beans, train staff, fair prices: street becomes coffee destination (Stag)
- Or cut prices, save on quality, steal customers today (Hare)
- If one commits to quality while other defects, committed one bears higher costs
- Both hesitate, waiting for signal of trust
- Challenge: assurance, not conflict
- Quality (Stag), Cut Costs (Hare)

Payoff Matrix (Monthly Profit in Rs):

Ramesh \ Suresh	**Stag**	**Hare**
Stag	**(80k, 80k)**	(30k, 55k)
Hare	(55k, 30k)	**(45k, 45k)**

Analysis Steps:

- **Two Nash Equilibria:**
  - (Quality, Quality): Superior outcome (80K each)
  - (Cut Costs, Cut Costs): Safe but inferior (45K each)
- **The Problem:** Quality only works if BOTH commit
- **Risk:** If you choose Quality and other defects, you get only 30K
- **Solution:** Build trust, pre-commitment, repeated interaction
- **Key Insight:** About mutual assurance, not defeating opponent

Ultimatum Game: Fairness Over Profit

- **Introduction**: People will sacrifice their own gain to punish perceived unfairness. This game demonstrates that humans are emotional moralists, not purely rational profit-maximisers, destroying the "Rational Man" economic model.
- **Situation**: Any offer/rejection setting: wage negotiations, settlement offers, supplier contracts, brand boycotts. Whenever perceived fairness can override strict economic logic.
- **Steps**:
  - Frame offers so they appear fair, not just profitable to you
  - Use anchoring to set fairness expectations early

- Give the other party a sense of agency in the process
  - Avoid visibly large asymmetries in payoffs
  - Expect rejection of “logically reasonable” offers that feel insulting
- **Examples**: Labour strikes rejecting technically positive wage offers; restaurant tipping even when you will never return; brand boycotts over perceived exploitation.

Ultimatum Game: Ramesh-Suresh Profit Split

The Scenario:

- Joint catering contract worth Rs 100,000
- Ramesh (proposer) must split profit with Suresh (responder)
- Ramesh offers a split
- Suresh can accept (both get their shares) or reject (both get zero)
- Rationally, any positive amount better than zero
- But fairness matters, people reject unfair offers

Possible Outcomes:

Ramesh's Offer	If Accepted	If Rejected
90K:10K split	(90k, 10k)	(0, 0)
70K:30K split	(70k, 30k)	(0, 0)
50K:50K split	(50k, 50k)	(0, 0)

Analysis:

- **Rational Prediction:** Suresh should accept any positive amount (even Rs 1)
- **Reality:** People reject offers below 20-30%
- **Fairness Norm:** Most offers around 40-50%
- **Why:** Fairness valued over pure profit
- **Key Insight:** Humans aren't purely profit-maximizing, social norms and emotions matter
- **Strategic Lesson:** Leave enough on table for others

Battle of the Sexes: Ramesh-Suresh Bakery Partnership

The Scenario:

- Both want “Breakfast & Coffee” combo with local bakery
- Ramesh wants bakery to use his premium beans for morning catering
- Suresh wants bakery to provide exclusive “Suresh-branded” croissants
- Both want partnership, both want cooperation
- Just can't agree on terms
- Struggle: who blinks first and accepts second-best choice?

Payoff Matrix (Monthly Additions):

Ramesh \ Suresh	Ramesh's Terms	Suresh's Terms
Ramesh's Terms	(0, 0)	(30k, 50k)
Suresh's Terms	(50k, 30k)	(0, 0)

Analysis Steps:

- **Two Nash Equilibria:**
  - (Suresh's, Ramesh's): Ramesh gets less but agrees
  - (Ramesh's, Suresh's): Suresh gets less but agrees
- **Key Feature:** Both prefer coordinating on EITHER outcome over no deal (0, 0)
- **But:** Each prefers different coordination point
- **Power:** With one who cares less, appears more stubborn, or commits first
- **Solutions:** Turn-taking, side payments, pre-commitment

Public Goods Game: The Free-Rider Problem

- **Introduction**: A public good is non-excludable and non-rivalrous. The free-rider problem emerges when individuals consume more than their fair share while paying less than their fair share, destroying the commons.
- **Situation**: Wikipedia donations, national defence, clean air, office coffee funds, open-source software maintenance, and voluntary neighbourhood upkeep.
- **Steps**:
  - Identify whether the good is genuinely non-excludable
  - Introduce social norms or visibility (people contribute more when observed)
  - Use conditional cooperation mechanisms (“I give if others give”)
  - Create excludability where possible (paywalls, memberships)
  - Regulate via taxation or mandated contribution
- **Examples**: Wikipedia donation drives; national vaccination campaigns; office kitchen cleanliness; funding public broadcasting.

Public Goods: Ramesh-Suresh Street Lighting

The Scenario:

- Street needs better lighting, costs Rs 20,000
- Each can contribute Rs 10,000 or nothing
- If installed, BOTH benefit (Rs 15,000 value each)
- If one contributes alone, they pay Rs 10K, get Rs 15K benefit = net Rs 5K
- If both contribute, each pays Rs 10K, gets Rs 15K = net Rs 5K each
- Temptation: let other pay, enjoy benefit for free

Payoff Matrix (Net Benefit in Rs):

Ramesh \ Suresh	Contribute	Free-Ride
Contribute	(5k, 5k)	(-10k, 15k)
Free-Ride	(15k, -10k)	**(0, 0)**

Analysis:

- **Dominant Strategy:** Free-ride for both
- **Nash Equilibrium:** (Free-Ride, Free-Ride) = (0, 0)
- **Best Outcome:** Both contribute = (5K, 5K)
- **The Problem:** Individual rationality leads to no public good
- **Solutions:** Mandatory contributions, reputation, repeated interaction, social norms

Matching Pennies: Pure Unpredictability

- **Introduction**: A purely competitive zero-sum game with no stable pure strategy. Your only winning approach is randomisation, if you have a pattern, the opponent exploits it.
- **Situation**: Penalty kicks; tax audit selection; security screening; military patrol routes; marketing launch timing against competitors, anywhere your adversary benefits from predicting you.
- **Steps**:
  - Identify that you are in a purely competitive, no-equilibrium-in-pure-strategies game
  - Calculate the optimal mixed strategy probabilities
  - Introduce genuine randomisation (dice, random number generators)
  - Resist the temptation to use “gut feel” patterns
  - Monitor whether the opponent is exploiting any pattern and adjust
- **Examples**: Penalty kick direction; customs inspection selection; submarine evasion routes; Rock-Paper-Scissors.

Matching Pennies: Ramesh-Suresh Timing Game

The Scenario:

- Ramesh and Suresh compete for morning rush customers
- Each must choose: Open Early (6 AM) or Normal (8 AM)
- Ramesh wins if choices match (both early or both normal)
- Suresh wins if choices differ (one early, one normal)
- Pure conflict, no cooperation possible
- Predictability is dangerous

Payoff Matrix (Customer Advantage):

Ramesh \ Suresh	Early	Normal
Early	(1, -1)	(-1, 1)
Normal	(-1, 1)	(1, -1)

Analysis:

- **No Pure Nash Equilibrium:** Always incentive to deviate
- **Solution:** Mixed strategy, randomize choices
- **Optimal:** Each plays Early 50%, Normal 50%
- **Key Insight:** Unpredictability is strategic
- **Examples:** Penalty kicks, rock-paper-scissors, military tactics
- **Lesson:** When opponent can exploit patterns, randomize

Cooperative Game Theory: Sharing the Gains

- **Introduction**: Cooperative game theory asks: when players can form coalitions and make binding agreements, how should gains be fairly and stably divided? The Shapley Value and the Core are the key solution concepts.
- **Situation**: Startup profit-sharing, political coalition formation, airline alliances, pharmaceutical joint ventures, supply-chain cost allocation, international carbon agreements.
- **Steps**:
  - Identify all possible coalitions and their total values
  - Calculate each player's marginal contribution (Shapley Value)
  - Check if the allocation is in the Core, no sub-coalition wants to break away
  - Negotiate side-payments if the allocation is outside the core
  - Build binding agreements and enforcement mechanisms
- **Examples**: Law firm profit splits; coalition government ministry allocations; airline revenue-sharing alliances; joint-venture R&D cost-sharing.

Cooperative Game Theory: Ramesh-Suresh Joint Venture

The Scenario:

- Both can bid jointly for corporate catering contract
- Alone: Ramesh can earn Rs 100K, Suresh Rs 80K
- Together: Combined capacity = Rs 250K total
- Question: How to split fairly?
- Shapley Value: Based on marginal contribution
- The Core: Allocations where no one wants to leave

Value Analysis:

Coalition	Total Value
Ramesh alone	Rs 100K
Suresh alone	Rs 80K
Both together	Rs 250K
Surplus from cooperation	**Rs 70K**

Fair Split Approaches:

- **Shapley Value:**
  - Ramesh: Rs 135K
  - Suresh: Rs 115K
- **Must be in Core:**
  - Ramesh gets $\geq$ Rs 100K
  - Suresh gets $\geq$ Rs 80K
  - Sum = Rs 250K
- **Key Insight:** Fair split considers outside options and marginal contributions
- **Applications:** Startups, coalitions, partnerships

Backward Induction: Think Backwards, Act Forward

- **Introduction**: To solve a sequential game, examine the final stage, find the optimal move there, then roll back logic to the start. It prevents moves today that create traps tomorrow.
- **Situation**: Chess endgames, business exit strategy design, career planning, negotiations with defined endpoints, syllabus design, retirement planning.
- **Steps**:
  - Define the final desired outcome clearly
  - Work backwards: what must be true one step before the end?
  - Continue rolling back, mapping each decision node
  - Identify the first move that leads to the desired end
  - Check for credibility, are all intermediate steps actually incentive-compatible?
- **Examples**: Chess endgame analysis; Indian parents planning children's education backwards from career goal; startup exit strategy mapped from IPO back to MVP.

Backward Induction: Ramesh Plans His Exit

The Scenario:

- Ramesh plans to sell his shop in 3 years
- Starts by imagining: What will buyer want?
- Answer: Stable brand, loyal customers, healthy margins
- Works backward: Price war today would create discount-chasing customers
- Brand would be worthless at exit
- So today's move: Raise quality, redesign space, position as premium

Decision Tree (Simplified):

Today	In 3 Years	Sale Value
Price War	Discount	Low (Rs 10L)
Premium Quality	Loyal high-value	High (Rs 50L)

Steps:

- **Define End Goal:** Sell for maximum value
- **Work Backward:** What creates value at exit? Loyal premium customers
- **One Step Before:** What builds loyalty? Consistent quality
- **Earlier Still:** What enables quality? Investment today
- **First Move:** Raise quality now, even if some customers leave
- **Key Insight:** Today's "loss" is tomorrow's setup

Forward Induction: Reading Hidden Intentions

- **Introduction**: Past actions reveal future intent. If a player makes a move that would be irrational *unless* they held a specific strong position, you can infer that strong position.
- **Situation**: Interpreting competitor signals in markets, reading investor confidence from founder behaviour, recruiting and interviewing, diplomacy, strategic partnership evaluation.
- **Steps**:
  - Observe a costly or unusual action by the other party
  - Ask: what beliefs would make this action rational?
  - Update your assessment of their type or intention
  - Adjust your strategy accordingly
  - Use the same logic to send credible signals about your own type
- **Examples**: A founder taking a Rs. 1 salary signals supreme confidence to investors; large Super Bowl ad spend signals financial strength; university degrees signal perseverance.

Forward Induction: Ramesh's Price Freeze Signal

The Scenario:

- Ramesh announces: Prices unchanged for 3 years, despite rising costs
- Looks reckless to Suresh
- But Ramesh is burning flexibility to signal confidence
- If Ramesh were unsure about footfall, this would box him into losses
- The fact he accepts risk suggests hidden strength
- Maybe superior location, loyal base, or cost advantage

Suresh's Inference:

Ramesh's Move	What It Signals
Price freeze	Confident in volume
Accepts margin pressure	Has cost advantage
Public commitment	Won't back down

Analysis Steps:

- **Observe Costly Action:** Price freeze with rising costs
- **Ask:** What beliefs make this rational?
- **Answer:** Confidence in strong customer base
- **Infer Type:** Ramesh likely has advantage
- **Update Strategy:** Don't compete on price, differentiate elsewhere
- **Key Insight:** Actions speak louder than words, costly signals are credible

Sequential Games: Turn-Based Strategy

- **Introduction**: In sequential games, players move one after another and can observe prior moves. Time and order of moves determine power, first movers set the tone, last movers often control the outcome.
- **Situation**: Real estate negotiation, chess, lobbying, wage bargaining, courtroom strategy, Stackelberg competition in duopoly markets.
- **Steps**:
  - Map the game tree with all decision nodes
  - Apply backward induction
  - Identify who benefits from moving first vs. last
  - Consider pre-commitment to shape subsequent play
  - Use observable actions to constrain the opponent's future choices
- **Examples**: House purchase negotiation (offer → counter → walk away → call back); Stackelberg leader in output competition; chess opening theory.

Sequential Games: Ramesh-Suresh Second Floor

The Scenario:

- Move 1: Ramesh adds second floor for remote workers
- Move 2: Suresh observes, then introduces quiet zone + faster WiFi
- Move 3: Ramesh notices noise complaints, adds premium pricing
- Move 4: Suresh offers membership passes with unlimited refills
- Move 5: Ramesh pivots to evening networking events
- Move 6: Suresh extends hours, markets as all-day workspace

Key Feature: Observable Moves

Each player sees previous moves before deciding

Strategic Advantages:

- **First Mover:**
  - Frames the game
  - Signals commitment
  - Sets agenda
- **Second Mover:**
  - Gains information
  - Can tailor response
  - Often controls outcome
- **Key Tools:** Backward induction, commitment, timing
- **Real Examples:** Negotiations, lobbying, chess

Simultaneous Games: Deciding Under Uncertainty

- **Introduction**: Players act at the same time without observing each other's choices. Strategy must be based on probability and beliefs about opponent rationality, not observed moves.
- **Situation**: Sealed tender auctions, stock market investing, general elections, simultaneous product launches, penalty kicks where the decision is truly simultaneous.
- **Steps**:
  - Enumerate all strategies and build a payoff matrix
  - Search for dominant strategies
  - If none exist, find Nash Equilibrium in pure strategies
  - If none in pure strategies, compute mixed strategy Nash Equilibrium
  - Use prior information about opponent types to refine predictions
- **Examples**: Sealed government contract bids; buying stocks before earnings announcements; Rock-Paper-Scissors.

Simultaneous Games: Ramesh-Suresh Menu Launch

The Scenario:

- Both planning to launch new menu items this month
- Must decide without knowing other's choice
- Options: Italian (pasta/pizza) or Asian (noodles/rice)
- If different: Each captures niche market
- If same: Split market, higher competition
- Cannot observe each other's prep until launch day

Payoff Matrix (Additional Revenue/m):

Ramesh \ Suresh	Italian	Asian
Italian	(20k, 20k)	(40k, 40k)
Asian	(40k, 40k)	(20k, 20k)

Analysis:

- **No Dominant Strategy:** Best choice depends on opponent
- **Two Nash Equilibria:** (Italian, Asian) and (Asian, Italian)
- **Challenge:** How to coordinate without communication?
- **Solutions:**
  - Past patterns
  - Focal points
  - Market research
  - Mixed strategies
- **Key Difference:** Must decide blindly, no info about opponent's move

Subgame Perfect Equilibrium: Credible Plans

- **Introduction**: Nash equilibrium can include non-credible threats that are never tested. Subgame perfection demands that your strategy must be optimal at *every* decision point, even in branches of the game tree you hope never to reach.
- **Situation**: Labor negotiations, military deterrence, corporate takeover defence, parent-child discipline, contract enforcement, anywhere threats or promises must be believed.
- **Steps**:
  - Draw the full game tree
  - Apply backward induction at every node
  - Remove strategies that are irrational at any sub-game
  - Ask: if this branch were reached, would I actually follow through?
  - If not, redesign your commitment to make it credible
- **Examples**: Parent who always carries through on stated consequences; proportional military response doctrines; poison-pill takeover defences that are actually executable.

Minimax Strategy: Minimising Maximum Loss

- **Introduction**: A conservative strategy that looks at the worst-case scenario of every decision and selects the one with the least-bad maximum loss. It is playing not to lose rather than playing to win.
- **Situation**: Risk management, insurance, cybersecurity, crisis planning, healthcare protocols, wherever catastrophic downside must be avoided even at cost of upside.
- **Steps**:
  - List all your strategies and all opponent strategies
  - Find the worst payoff for each of your strategies (the "minimum")
  - Select the strategy with the highest minimum payoff
  - Compare minimax cost against expected value analysis
  - Adjust based on risk tolerance and reversibility of outcomes
- **Examples**: Travel insurance; diversified investment portfolios; pre-nuptial agreements; cybersecurity breach response planning.

Bayesian Games: Decisions with Incomplete Information

- **Introduction**: Most real-world games involve uncertainty about opponents' types, payoffs, or capabilities. Players form beliefs, observe actions, update via Bayes' rule, and choose strategies maximising expected payoffs given current beliefs.
- **Situation**: Used car markets, job interviews, insurance underwriting, military intelligence assessment, product demand forecasting, dating and relationship formation.
- **Steps**:
  - Assign prior probability distributions over opponent types
  - Identify what signals or actions reveal information
  - Update beliefs using Bayes' Rule as evidence arrives
  - Compute expected payoffs across opponent types
  - Choose strategy maximising expected payoff; recognise your actions also signal your type
- **Examples**: Employer updating beliefs about candidate quality through interview; used-car buyer reading seller signals; poker player updating hand estimate from betting patterns.

Strategic Commitment: Limiting Your Own Options

- **Introduction**: Paradoxically, you can increase bargaining power by *reducing* your own options. If you credibly cannot back down, the other party must adapt to your position.
- **Situation**: Wage negotiations, competitive market entry threats, diplomatic standoffs, self-control problems, anywhere restricting future flexibility strengthens your current position.
- **Steps**:
  - Identify the decision you want the other party to take
  - Find a way to make your own retreat genuinely costly or impossible
  - Make the commitment visible and verifiable
  - Ensure the commitment is specific, not vague
  - Time the commitment before negotiations begin
- **Examples**: Cortés burning his ships; automatic pension deductions; publicly announced deadlines; contracts with heavy penalty clauses; hunger strikes.

Strategic Commitment: Suresh's Organic Contract

The Scenario:

- Suresh tired of endless price wars draining margins
- Signs 5-year non-breakable contract with organic bean supplier
- Locks into higher input costs
- Frames contract copy near counter where all can see
- Message to Ramesh: Pricing at Rs 50 now impossible
- Removed own escape route to gain leverage

Before vs After Commitment:

Scenario	Before	After
Can lower prices?	Yes	No (contract)
Ramesh's threat	Credible	Not credible
Price war	Continues	Ends

Analysis:

- **Paradox:** Strength from constraint
- **By giving up flexibility:** Gains influence
- **Key Requirements:**
  - Visible (framed contract)
  - Credible (legally binding)
  - Costly to reverse
- **Outcome:** Ramesh stops price war, focuses on differentiation
- **Lesson:** Your lack of options becomes their problem

Pre-Commitment Devices: Binding Your Future Self

- **Introduction**: We are often two people, the planner and the impulsive doer. Pre-commitment devices are strategies the planner uses to constrain the doer before temptation strikes.
- **Situation**: Dieting, saving, avoiding addictions, maintaining study habits, managing social media overuse, any area where your future self's impulsive decisions conflict with your present-self's goals.
- **Steps**:
  - Identify the specific moment of temptation
  - Introduce a barrier before that moment (not after)
  - Make the barrier automatic, not dependent on future willpower
  - Increase the cost of overriding the commitment
  - Use social accountability to amplify the pre-commitment
- **Examples**: Odysseus tying himself to the mast; deleting social media apps; freezing credit cards in ice; gym buddies; automatic savings transfers.

Credible Threats: Making Promises Believable

- **Introduction**: A threat is only useful when it is credible. If carrying out the threat hurts you more than the opponent, it will be called as a bluff. The incentive structure must make follow-through rational.
- **Situation**: Parental discipline, deterrence strategy, price-match guarantees, union strike threats, contractual penalty clauses, wherever compliance requires the other party to believe punishment is real.
- **Steps**:
  - Ask honestly: would I actually carry this out?
  - Adjust the threat so it costs you less than compliance costs the opponent
  - Make the threat automatic (remove discretion)
  - Build a track record of following through
  - Communicate the threat early and unambiguously
- **Examples**: Nuclear deterrence structures; price-match guarantees (credible threat to rivals); central bank inflation targets; unconditional resignation letters.

Brinkmanship: Playing with Fire

- **Introduction**: Deliberately pushing a situation to the edge of mutual disaster, or creating the appearance of losing control, to frighten the opponent into conceding. The shared risk becomes the leverage.
- **Situation**: Debt-ceiling negotiations, strikes timed to peak season, geopolitical confrontations, diplomatic ultimatums, late-stage acquisition negotiations.
- **Steps**:
  - Assess the cost of mutual disaster, it must be roughly equal or worse for the opponent
  - Create visible risk of escalation (not just the threat)
  - Maintain plausible deniability, "I may not be able to stop this."
  - Keep an offramp for the opponent to retreat without total humiliation
  - Know precisely when to de-escalate
- **Examples**: Indo-Pak border tensions; union strikes during harvest season; debt-ceiling confrontations; Cold War nuclear postures.

Burning Bridges: Eliminating Retreat

- **Introduction**: By deliberately eliminating your own escape routes, you signal unshakeable commitment. When retreat is impossible, threats become credible, bluffs become real, and opponents must take your position seriously.
- **Situation**: Market entry with irreversible investment, public goal announcements, constitutional lock-ins, military positions, start-up pivots where the old product is shut down.
- **Steps**:
  - Identify what escape routes the opponent expects you to take
  - Eliminate those routes publicly and irreversibly
  - Signal the elimination to your opponent clearly
  - Ensure the commitment is specific enough to be believed
  - Carefully weigh: are you certain enough to make retreat impossible?
- **Examples**: Cortés burning ships in Mexico; no-refund advance-payment policies; publicly announced deadlines; constitutional supermajority requirements; military forces holding indefensible positions.

Reputation Effects: Your History Precedes You

- **Introduction**: In repeated interactions, your past actions determine your future payoffs. A reputation for being "tough" or "honest" changes how others play against you, it is a capital asset that earns compound returns.
- **Situation**: Brand management, credit markets, diplomatic relations, business partnerships, personal professional networks, any context with repeated interaction and observable history.
- **Steps**:
  - Identify what reputation you want to build
  - Make short-term sacrifices to establish credibility
  - Ensure your actions are observable

- Maintain consistency, reputation is destroyed by exceptions
  - Invest in platforms where reputation is recorded (ratings, testimonials, public records)
- **Examples**: Tata Group's century-long reputation; credit scores; eBay seller ratings; central bank inflation credibility; Mafia omertà.

Escalation of Commitment: The Sunk Cost Trap

- **Introduction**: The irrational decision to continue a failing course of action because of past investments. Rational strategy demands looking only forward, past costs are gone and irrelevant to future decisions.
- **Situation**: Ongoing projects with poor ROI, failing relationships, bad stocks held too long, unfinished degrees, prolonged wars, any situation where sunk cost drives continued investment.
- **Steps**:
  - Separate past costs from future costs and benefits
  - Ask: if I were starting today, would I make this investment?
  - Establish clear kill-criteria in advance
  - Assign an independent reviewer not emotionally attached to the project
  - Reframe exit as strategic intelligence, not failure
- **Examples**: The Vietnam War; waiting for a bus already 30 minutes late; gambling to "win it back"; keeping a failing business unit alive.

Tit-for-Tat: Reciprocity Works

- **Introduction**: The winning strategy in repeated Prisoner's Dilemma tournaments: cooperate first, then mirror whatever the opponent did last round. Simple, forgiving, but retaliatory, the algorithm of human morality.
- **Situation**: Trade negotiations, business partnerships, gifting cultures, arms control agreements, neighbourhood relations, any long-term repeated interaction where defection must be deterred.
- **Steps**:
  - Open by cooperating unconditionally
  - Immediately mirror any defection with one round of defection
  - Return to cooperation the moment the opponent cooperates
  - Never defect first
  - Communicate the strategy so the opponent understands the pattern
- **Examples**: Trade tariff reciprocity; reciprocal dinner invitations; ceasefire agreements conditioned on compliance; eye-for-an-eye justice systems.

Tit-for-Tat: Ramesh-Suresh Reciprocity

The Scenario:

- Both trying to undercut with lower prices, aggressive promotions
- Pattern: Aggression met with aggression, both lose margins
- Ramesh tries different: Offers free pastry with morning coffee
- Suresh notices, responds with small gesture
- Respect boundaries, retaliate only when needed, forgive small moves
- Coffee war → cooperation w tit-for-tat

Tit-for-Tat Strategy:

Step	Action
1. First move	Cooperate (be nice first)
2. If opponent cooperates	Continue cooperating
3. If opponent defects	Retaliate once (proportional)
4. If opponent returns	Forgive, resume cooperation

Why It Works:

- **Nice:** Never defects first
- **Retaliatory:** Punishes defection immediately
- **Forgiving:** Returns to cooperation
- **Clear:** Opponent understands pattern
- **Examples:**
  - Trade tariffs
  - Social invitations
  - Ceasefire agreements
- **Key Insight:** Golden Rule with teeth, cooperation can survive among self-interested agents

Shadow of the Future: Long-Term Thinking

- **Introduction**: Cooperation is sustainable only when the "shadow of the future" looms large. If interactions repeat indefinitely, the long-term reputational loss from cheating outweighs the short-term gain.
- **Situation**: Family businesses, small-town communities, subscription models, diplomatic relations, supplier relationships, wherever continuity of interaction creates accountability.
- **Steps**:
  - Increase the probability and salience of future interactions
  - Reduce the discount rate on future payoffs
  - Establish visible accountability systems
  - Build reputational stakes into short-term interactions
  - Create switching costs that make exit from the relationship costly
- **Examples**: Not cheating your local vendor because you see them daily; subscription loyalty programs; diplomatic relations sustained by future needs; small-town honesty norms.

Signalling: Sending Messages Through Action

- **Introduction**: When one party holds private information, they must use observable actions to convey it credibly. A good signal must be both observable and difficult to fake, otherwise it conveys nothing.
- **Situation**: Job market signalling, brand advertising, corporate dividend policy, product warranties, personal reputation management, any information-asymmetric interaction.
- **Steps**:
  - Identify the hidden information you need to convey
  - Find an action that is cheap for you (high type) but expensive for a low type to mimic
  - Make the signal observable
  - Maintain signal consistency over time
  - Verify that recipients correctly interpret the signal
- **Examples**: Wedding rings as availability signals; dividends signalling corporate financial health; peacock tails as evolutionary fitness signals; product warranties signalling quality.

Costly Signalling: Credibility Through Sacrifice

- **Introduction**: Talk is cheap; for a signal to be trusted in high-stakes games it must be too expensive for a "faker" to afford. The cost *is* the proof.
- **Situation**: High-stakes relationship formation, luxury brand signalling, investor confidence building, gang initiation rites, religious fasting, any context where cheating would be profitable unless signalling is costly.
- **Steps**:
  - Identify what claim you need others to believe
  - Find a costly action that only someone with the genuine claim would rationally take
  - Ensure the cost is observable and verified
  - Calibrate cost: high enough to deter fakers, not so high it deters honest signalers
  - Repeat over time to build a track record
- **Examples**: Expensive engagement rings; charitable donations by the wealthy; fraternity initiation costs; luxury goods as wealth signals; religious fasting as community commitment.

Rational Irrationality: Strategy Through Unpredictability

- **Introduction**: Being perfectly rational makes you predictable and exploitable. Feigning, or genuinely being, irrational alters the opponent's payoff matrix, forcing them to yield to avoid an unpredictable catastrophe.
- **Situation**: Geopolitical standoffs, extreme negotiation leverage, crowd control, deterring bullying, anywhere the opponent's certainty about your behaviour is a weapon against you.
- **Steps**:
  - Identify where your predictability is being exploited
  - Introduce deliberate unpredictability or "madman" signals
  - Ensure the irrationality is credible, genuine emotional investment helps
  - Have a private off-switch you control even while appearing to lack one
  - Return to rational behaviour once the leverage has worked
- **Examples**: Nixon's "Madman Theory" in Vietnam; protester self-immolation as commitment device; doomsday devices in deterrence; road rage escalating beyond rational limits.

First-Mover Advantage: Being Early Pays

- **Introduction**: The first player to enter a market can lock in resources, customers, and standards. They set the rules that followers must accept. The brand can become the category name itself.
- **Situation**: New technology categories, emerging markets, standard-setting battles, patent races, geographic expansion into underpenetrated regions.
- **Steps**:
  - Identify the untapped market and timing window
  - Move decisively before the window closes
  - Use early position to lock in key resources (talent, patents, partners)
  - Set the standard others must comply with
  - Invest early in switching costs and network effects
- **Examples**: Xerox defining "photocopying"; Amazon in e-commerce; Coca-Cola's global brand position; Bitcoin as first cryptocurrency; squatters' rights laws.

Second-Mover Advantage: Learning from Mistakes

- **Introduction**: Being first is risky and expensive. Second movers free-ride on the pioneer's investments, learn from their errors, and adopt superior technology once the market is de-risked.
- **Situation**: Technology markets where innovation moves rapidly, emerging product categories after initial market education, regulatory environments that reward fast followers over pioneers.
- **Steps**:
  - Monitor pioneers carefully; let them educate the market
  - Identify pioneer weaknesses and customer dissatisfactions
  - Build a superior product on validated demand
  - Strike at the moment of pioneer fatigue or overextension
  - Use scale and capital efficiency to leapfrog
- **Examples**: Google after Yahoo; Facebook after MySpace; Samsung after Apple; generic drugs after patent expiry; Zara replicating runway fashion rapidly.

Entry Deterrence: Protecting Market Territory

- **Introduction**: Incumbents can make it structurally unprofitable for new players to enter by over-investing in capacity, pricing aggressively, or controlling key complements, signalling "don't bother."
- **Situation**: Market leaders facing potential competition, platform companies with network effects, utilities and infrastructure businesses, industries with high capital requirements.
- **Steps**:
  - Signal excess capacity credibly to deter entry
  - Use loyalty programs and switching costs to lock in customers
  - Acquire critical input suppliers or distribution channels
  - Use limit pricing to keep post-entry profits unattractive
  - File strategic patents around core technology
- **Examples**: Jio's free-data launch; frequent-flyer mile programmes; patent thickets in pharmaceuticals; exclusive distributor contracts; Amazon's low-margin saturation strategy.

Limit Pricing: Discouraging Competition Through Price

- **Introduction**: Setting prices lower than monopoly level, but above competitive cost, specifically to make market entry unprofitable for potential rivals without sacrificing all incumbent profits.
- **Situation**: Dominant firms in markets with moderate entry barriers, commodity markets with cost-advantage incumbents, digital platforms with significant scale economics.
- **Steps**:
  - Estimate the entrant's cost structure
  - Set price just below the level that makes entry profitable for them
  - Ensure price is above your own average cost
  - Signal that price will stay low even post-entry
  - Back the signal with visible cost advantages
- **Examples**: OPEC pricing decisions; Microsoft Windows pricing in the 1990s; India's telecom data pricing wars; Amazon Basics undercutting third-party sellers.

Strategic Alliances: Temporary Partnerships

- **Introduction**: Cooperation between competitors, the "frenemy" relationship, to achieve specific goals like scale or access. Partners cooperate on the platform while competing on the product.
- **Situation**: Airline codeshare agreements, technology standards bodies, joint R&D ventures, political coalitions, supply-chain partnerships with horizontal competitors.
- **Steps**:
  - Identify the specific shared objective that justifies cooperation
  - Bound the alliance scope, cooperate only where interests align
  - Establish clear governance and exit mechanisms
  - Protect proprietary information carefully
  - Plan for the alliance ending when objectives diverge
- **Examples**: NATO; airline codeshare agreements; Spotify and Uber; Apple using Samsung screens; pharmaceutical joint ventures for R&D.

Cheap Talk: Words Without Bite

- **Introduction**: Costless communication that does not directly affect payoffs. Because it is free, it is often misleading. But when interests align, cheap talk can successfully coordinate without commitment.
- **Situation**: Dating app profiles, political campaign promises, CEO letters, advertising claims, diplomatic statements, anywhere words are free and verification is costly.
- **Steps**:
  - Distinguish cheap talk from costly signalling
  - Ask: does the speaker have incentives to lie? If yes, discount heavily
  - Look for corroborating actions, not just words
  - Use cheap talk to coordinate when interests align
  - Design systems that make lying costly to convert cheap talk to credible talk
- **Examples**: Street vendor's "final offer"; political manifesto promises; CEO shareholder letters; football pre-match trash talk.

Perfect vs. Imperfect Information

- **Introduction**: Perfect information means all players see all moves and the full game state (chess). Imperfect information means some moves are hidden (poker). This single distinction transforms strategy from calculation to psychology.
- **Situation**: Distinguishing which type of game you are in determines whether to invest in deeper analysis vs. psychological profiling and information management.
- **Steps**:
  - Determine what information is observable vs. hidden
  - In perfect information games, use backward induction and deep calculation
  - In imperfect information games, model opponent types and beliefs
  - Invest in revealing opponent information through screening
  - Conceal your own information strategically
- **Examples**: Chess (perfect) vs. Poker (imperfect); transparent vs. sealed-bid auctions; open parliamentary debates vs. backroom diplomacy.

Common Knowledge: Synchronised Understanding

- **Introduction**: Mutual knowledge means everyone knows X. Common knowledge means everyone knows that everyone knows X, to infinite recursion. This transition from private belief to shared awareness enables coordinated mass action.
- **Situation**: Social movements, bank runs, market bubbles, cultural paradigm shifts, anywhere private individual beliefs must become public to trigger collective behaviour change.
- **Steps**:
  - Identify information that is privately held but widely shared
  - Find a public event that makes the private knowledge common
  - Anticipate the coordinated behaviour change that follows
  - Use or create common knowledge to coordinate allies
  - Suppress common knowledge formation to prevent unwanted coordination
- **Examples**: Arab Spring via social media; bank runs on shared panic; #MeToo making isolated experiences collective; market bubbles bursting on shared doubt.

Information Asymmetry: The Knowledge Gap

- **Introduction**: When one party has more or better information than the other, the informed party can exploit the gap. This imbalance causes market failures as uninformed parties fear being cheated and exit.
- **Situation**: Doctor-patient, mechanic-driver, real estate agent-buyer, insurer-insured, employer-employee relationships, any principal-agent context with unequal information.
- **Steps**:
  - Identify who holds the information advantage
  - If uninformed, seek signals, warranties, or third-party verification
  - If informed, decide whether to disclose (building trust) or withhold (exploiting gap)
  - Design mechanisms that align the informed party's incentives with disclosure
  - Reduce asymmetry through transparency norms or regulation
- **Examples**: Mechanics exploiting car-ignorant customers; insider trading; real estate agents prioritising commission over client interest.

Adverse Selection: Hidden Types Cause Problems

- **Introduction**: When quality is hidden, bad options (lemons) drive out good ones. Markets collapse because high-quality types leave rather than subsidise low-quality types, leaving only the worst in the pool.
- **Situation**: Insurance markets, used-car markets, online dating, volunteer armies, all-you-can-eat pricing, any market where participants self-select based on private information.
- **Steps**:
  - Identify the hidden characteristic causing selection
  - Design screening mechanisms to sort types
  - Use signalling to allow high types to distinguish themselves
  - Price discriminate based on revealed preference
  - Introduce mandatory pooling (e.g., compulsory insurance) to prevent market unravelling
- **Examples**: Akerlof's used-car "lemons" market; health insurance death spirals; all-you-can-eat buffets attracting the most voracious eaters.

Moral Hazard: Risk Without Responsibility

- **Introduction**: When a person is insulated from the consequences of their actions, they behave more recklessly than if they bore the full risk. Shielding from downside destroys incentives for care.
- **Situation**: "Too Big to Fail" banks, comprehensively insured drivers, rent-controlled tenants, CEOs with golden parachutes, any principal-agent setting where consequences are borne by a third party.
- **Steps**:
  - Identify who bears the cost of risky behaviour
  - Introduce co-payments, deductibles, or clawbacks to re-align incentives
  - Increase monitoring to observe hidden actions
  - Design contracts with performance-based compensation
  - Use reputational accountability to supplement formal incentives
- **Examples**: Banks gambling with taxpayer-backed deposits; drivers with comprehensive coverage taking more risks; CEOs with severance packages taking reckless bets.

Screening: Testing to Reveal Type

- **Introduction**: The uninformed party sets up a test that induces the informed party to reveal hidden information through self-selection. The design makes lying costly, so truth-telling becomes rational.
- **Situation**: Job interviews, insurance underwriting, university admissions, credit scoring, deductible-based insurance pricing, wherever the uninformed party needs to sort by hidden quality.
- **Steps**:
  - Identify the hidden type that matters (health, ability, commitment)
  - Design a menu of options where each type prefers the option intended for them
  - Ensure the "wrong" choice is costly enough to deter mimicry
  - Observe self-selection and update beliefs
  - Refine the screen as participants learn to game it
- **Examples**: Insurance deductibles revealing health risk; Saturday-night stay airline pricing separating business from leisure; coding tests in job interviews.

Bargaining Power: Influencing Expectations

- **Introduction**: Your power in any negotiation is determined not by your loudness but by your BATNA, Best Alternative to a Negotiated Agreement. The party who can most credibly walk away controls the table.
- **Situation**: Salary negotiations, merger talks, hostage negotiations, real estate deals, supplier contracts, union collective bargaining, all negotiations where alternatives determine leverage.
- **Steps**:
  - Identify and actively improve your BATNA before negotiating
  - Try to understand and weaken the other party's BATNA
  - Never reveal a weak BATNA
  - Signal your BATNA credibly through actions, not just words
  - Anchor the negotiation range early with your opening offer
- **Examples**: Having another job offer in salary negotiations; water vendor in a desert vs. one by a river; buyers in a buyer's market; monopolists with no alternatives facing them.

Nash Bargaining Solution: Fair Division

- **Introduction**: A mathematical approach to finding a "fair" bargaining outcome satisfying efficiency (nothing wasted), symmetry (equal bargainers get equal shares), and individual rationality. It creates a focal point for agreement.
- **Situation**: M&A pricing, climate burden-sharing agreements, royalty splits, plea bargaining, joint-venture profit allocation, wherever a principled division of surplus is needed.
- **Steps**:
  - Identify each party's outside option (disagreement payoff)
  - Calculate the total surplus from cooperation
  - Find the split that maximises the product of surplus gains for both parties
  - Check for symmetry adjustments based on relative patience or risk aversion
  - Use the solution as a benchmark to evaluate proposed deals
- **Examples**: M&A deal pricing relative to stand-alone values; climate agreement burden-sharing; management-union wage settlements avoiding strikes.

Rubinstein Bargaining Model: Patience Wins

- **Introduction**: When two parties alternate offers over time, each delay costs both sides as the pie shrinks. But the more patient party captures a larger share. With rational players, the first offer is accepted immediately, delay signals irrational commitment or private information.
- **Situation**: Real estate negotiations with carrying costs, labour strikes, divorce proceedings, acquisition talks with dwindling startup runway, international trade negotiations.
- **Steps**:
  - Assess your discount rate, how much does delay cost you?
  - Assess the opponent's discount rate, do they have burning runway?
  - Make your first offer close to the equilibrium solution (it should be accepted)
  - Signal patience credibly
  - Recognise that real delays indicate incomplete information or irrational commitment, adjust accordingly
- **Examples**: Car price negotiation as both parties' time costs mount; union strikes where both sides bleed daily; startup acquisition as funding runs out.

Rent-Seeking: Gaining Without Creating

- **Introduction**: Spending resources to increase one's share of existing wealth without creating new wealth. It is economically wasteful, consumes talent in zero-sum activities, and corrupts strategic incentives.
- **Situation**: Lobbying for regulations, patent trolling, occupational licensing cartels, corruption, feudalistic rent extraction, any activity focused on redistribution rather than value creation.
- **Steps**:
  - Distinguish between value-creating and rent-seeking activities in your strategy
  - Calculate the full social cost of lobbying vs. innovating
  - Identify rent-seeking by competitors and design regulation to block it
  - Avoid engaging in rent-seeking even when profitable, it invites retaliation and regulatory scrutiny
  - Advocate for institutional rules that reward production over position

- **Examples**: India's "Licence Raj" lobbying; patent trolls; occupational licensing cartels; corruption and bribery as market entry barriers.

Mechanism Design: Engineering the Rules

- **Introduction**: Reverse game theory, instead of predicting what players do given rules, we ask what rules to design to elicit desired behaviour. It is the architecture of incentives, making truth-telling the best strategy.
- **Situation**: School choice systems, spectrum auctions, kidney exchange, tax code design, medical residency matching, any context where the designer controls the rules but not the players.
- **Steps**:
  - Define the social objective (efficiency, fairness, revenue)
  - Identify the private information players hold
  - Design a mechanism where revealing true information maximises individual payoffs (incentive-compatibility)
  - Ensure participation is individually rational
  - Test for manipulation and iterate
- **Examples**: NRMP medical residency matching; FCC spectrum auctions; school choice algorithms; kidney exchange programmes; Vickrey second-price auctions.

Auction Theory: Strategic Bidding

- **Introduction**: Different auction formats (English, Dutch, sealed-bid, Vickrey) elicit different bidder behaviours and revenue outcomes. The winner's curse, overpaying because winning means everyone else bid lower, is the central hazard.
- **Situation**: IPL player auctions, government spectrum auctions, estate sales, eBay bidding, Google Ads, procurement auctions, privatisation of state assets.
- **Steps**:
  - Identify the auction format and its incentive structure
  - Estimate your true private value, resist the winner's curse by bidding below your estimate in common-value auctions
  - Observe competitor bidding patterns
  - Use the Revenue Equivalence Theorem to evaluate format choices
  - Bid truthfully in second-price (Vickrey) auctions, it is the dominant strategy
- **Examples**: IPL team auctions for player talent; Google AdSense second-price mechanism; eBay online bidding; government bond issuance.

Matching Markets: Stable Allocations

- **Introduction**: Some markets cannot use prices to clear allocation (kidney donation, medical residency). Matching algorithms create stable outcomes where no unmatched pair would both prefer each other, preventing market unravelling and black markets.
- **Situation**: Medical residency assignment, school choice, kidney exchange networks, entry-level labour markets, organ donation registries, college admissions.
- **Steps**:
  - Verify that prices cannot or should not clear the market
  - Gather complete preference lists from both sides
  - Apply the Gale-Shapley deferred-acceptance algorithm
  - Check that the resulting match is stable (no blocking pairs)
  - Design incentives so participants reveal true preferences
- **Examples**: NRMP medical residency (Nobel Prize-winning application); school choice in NYC and Boston; kidney exchange chains; entry-level lawyer placement.

Network Effects: Platforms and Lock-In

- **Introduction**: The value of a product increases with the number of its users. This creates winner-take-most dynamics where leading platforms grow larger while challengers die, not because the product is better but because the network is bigger.
- **Situation**: Social media, payment networks, operating systems, languages, stock exchanges, communication platforms, any product whose value scales with user count.
- **Steps**:
  - Measure same-side and cross-side network effects
  - Subsidise early adoption to reach critical mass
  - Design for interoperability or, strategically, against it
  - Use data network effects to reinforce technical advantages
  - Identify multi-homing costs, high multi-homing weakens lock-in
- **Examples**: WhatsApp's dominance despite imperfect product; Visa/Mastercard duopoly; Windows OS market; English as global business language.

Two-Sided Markets: Balancing Multiple Sides

- **Introduction**: Platforms facilitating interaction between two distinct groups face a chicken-and-egg bootstrapping problem. The strategy requires subsidising one side to attract the other, then extracting value from whichever side has inelastic demand.
- **Situation**: Ride-sharing, dating apps, gaming consoles, newspapers, credit cards, shopping malls, any platform requiring both supply and demand side to create value.
- **Steps**:
  - Identify which side is the "hard side" to attract
  - Subsidise the hard side heavily to build critical mass
  - Charge the side with the most to gain from the other's presence
  - Manage cross-side network effects, more of one side should benefit the other
  - Prevent disintermediation, keep both sides transacting on the platform
- **Examples**: Uber subsidising drivers and riders; gaming consoles subsidising hardware to attract developers; newspapers giving free content to attract advertisers.

Tragedy of the Commons: Overuse Destroys Resources

- **Introduction**: When a shared resource is unregulated, individuals consume it to exhaustion, the benefit of consumption is private, but the cost of depletion is borne collectively. Rational individuals destroy the commons.
- **Situation**: Overfishing, groundwater depletion, traffic congestion, air pollution, office kitchen misuse, climate change, any shared resource without governance.
- **Steps**:
  - Make the resource excludable (property rights or licensing)
  - Introduce quotas or Pigouvian taxes on usage
  - Build community norms and social monitoring

- Apply Ostrom's principles: local rules, graduated sanctions, collective governance
  - Create long-term stakes in the resource's health
- **Examples**: Delhi's smog from uncurbed driving; global overfishing; groundwater depletion in agricultural regions; climate change as a global commons failure.

Voting Paradoxes: When Democracy Fails

- **Introduction**: Arrow's Impossibility Theorem proves that no rank-order voting system can perfectly satisfy all criteria of fairness simultaneously. Every voting system can be gamed or produce inconsistent collective choices.
- **Situation**: Political elections with multiple candidates, boardroom decisions, committee voting, Oscar nominations, budget prioritisation, any collective choice aggregating individual preferences.
- **Steps**:
  - Understand the voting mechanism and its incentives
  - Identify potential spoiler effects and strategic voting incentives
  - Consider alternative mechanisms (ranked-choice, approval voting)
  - Account for agenda-setting power, who controls the order of votes
  - Design voting systems to minimise strategic manipulation given the context
- **Examples**: Spoiler candidates splitting the vote; gerrymandering; strategic "lesser-evil" voting; Condorcet cycles in committee decisions.

Volunteer's Dilemma: Who Acts First?

- **Introduction**: When a group needs one person to incur a cost for everyone's benefit, each member waits for another to volunteer. The larger the group, the more severe the problem, resulting in paralysis even when action is clearly needed.
- **Situation**: Emergency bystander responses, whistleblowing, group project initiation, social movements, bug reporting, standing ovations, reporting organisational wrongdoing.
- **Steps**:
  - Assign explicit responsibility to a specific person, diffused responsibility equals paralysis
  - Reward first-movers visibly and publicly
  - Reduce the cost of volunteering (anonymity, indemnification)
  - Break large groups into smaller accountable units
  - Build a culture where volunteering is the expected norm
- **Examples**: Bystander effect in the Kitty Genovese case; whistleblowers in corporations; group projects where no one starts; first protesters in social movements.

Focal Points (Schelling Points): Coordinating Without Communication

- **Introduction**: When multiple equilibria exist and communication is impossible, people converge on solutions that stand out through prominence, symmetry, or cultural salience. Self-fulfilling expectations create convergence without explicit agreement.
- **Situation**: Emergency rendezvous, social conventions, negotiation anchors, ceasefire boundary selection, standard pricing points, any coordination without the ability to communicate.
- **Steps**:
  - Identify which options stand out as "obvious" in context
  - Use round numbers and symmetric solutions as coordination anchors
  - In design, make your desired outcome the salient focal point
  - Use cultural and historical reference points to guide coordination
  - Recognise that your opponents will also search for the same focal point
- **Examples**: Grand Central Station at noon as a meeting point; round-number salary anchors (Rs. 10 lakh, not Rs. 9.83 lakh); ceasefire lines at rivers; splitting bills equally.

The Art of War and Diplomacy: Ancient Strategic Wisdom

- **Introduction**: Chanakya and Sun Tzu wrote millennia ago, yet CEOs quote them today. Because technology changes but human nature and conflict do not. The highest strategy is to win without fighting.
- **Situation**: Corporate competition, geopolitical diplomacy, organisational politics, competitive intelligence, brand wars, wherever perception management and asymmetric information can substitute for direct confrontation.
- **Steps**:
  - Know yourself and your enemy, assess strengths and weaknesses honestly
  - Win without fighting: use perception, reputation, and deception before force
  - Choose the battlefield, fight where you are strong, not where the enemy is
  - Use spies and intelligence before commitment
  - Strike at the opponent's strategy, not their army
- **Examples**: Corporate hostile takeover via proxy; soft power (Bollywood/Hollywood); guerrilla marketing; peace treaties as strategic positioning; cyber warfare.

Prospect Theory: Loss Aversion Changes Everything

- **Introduction**: Kahneman and Tversky's theory shows humans evaluate outcomes relative to a reference point, not absolute wealth, and losses hurt roughly twice as much as equivalent gains feel good. This reshapes all strategic interaction.
- **Situation**: Investor behaviour, labour negotiations, political framing, consumer pricing, legal settlements, any context where framing around a reference point shifts decisions away from rational maximisation.
- **Steps**:
  - Identify the reference point of your counterpart
  - Frame your offer as avoiding a loss rather than achieving a gain
  - Recognise the endowment effect, people overvalue what they already possess
  - Avoid triggering loss aversion by anchoring to a favourable reference point early
  - Exploit loss aversion in competitors while designing your own decisions around rational expected value
- **Examples**: Investors refusing to sell losing stocks; homeowners overpricing due to emotional attachment; unions rejecting wage cuts even to prevent layoffs; governments escalating losing wars to "not lose face."

Conclusion and Next Steps

The Infinite Game: Key Takeaways

What We Learned

- You are never playing solitaire, always chess
- Incentives trump intentions every time
- Information is the most valuable commodity
- Credibility must be earned through action
- Cooperation requires design, not hope
- Reputation compounds over decades
- The framing of a choice changes the choice

The Hierarchy of Strategic Wisdom

- Know which game you are in
- Know who the other players are
- Know their incentives
- Know your BATNA
- Make credible commitments
- Build reputation relentlessly
- **Play to keep playing**

The Ultimate Lesson
In the *Finite Game* of a football match, you play to win. In the *Infinite Game* of life, you play to **keep playing**. Stay adaptable.

Applying Game Theory: Your Action Plan

This Week

- Map one negotiation you are in as a game tree
- Identify your BATNA and improve it
- Find one Nash Equilibrium you are stuck in and ask: *what changes the game?*
- Spot one dominant strategy and use it

This Month

- Redesign one professional relationship using Tit-for-Tat
- Audit your reputation signals
- Find one rent-seeking activity and eliminate it
- Implement one pre-commitment device for a personal goal

Applying Game Theory: Your Action Plan

Ongoing Practice

- Before every significant decision: *who else is deciding, and what are their payoffs?*
- In every negotiation: *what is the other party's reference point?*
- In every conflict: *is this zero-sum or can I grow the pie?*
- In every commitment: *is this threat or promise actually credible?*

Remember
Game Theory is a map, not the territory. Real people are irrational, emotional, and surprising. Use these models as lenses, not algorithms.

Essential Reading: Resources to Go Deeper

Foundational Books

- **Thinking Strategically**, Dixit & Nalebuff (best accessible intro)
- **The Art of Strategy**, Dixit & Nalebuff (applied game theory)
- **Games of Strategy**, Dixit, Skeath & McAdams (textbook)
- **The Strategy of Conflict**, Thomas Schelling (focal points, brinkmanship)
- **Thinking, Fast and Slow**, Daniel Kahneman (Prospect Theory)
- **Getting to Yes**, Fisher & Ury (applied negotiation)

Advanced & Specialised

- **Arthashastra**, Kautilya (Chanakya), ancient strategy
- **The Art of War**, Sun Tzu, conflict & deception
- **Predictably Irrational**, Dan Ariely, behavioural economics
- **Co-opetition**, Brandenburger & Nalebuff, business strategy
- **The Infinite Game**, Simon Sinek, finite vs. infinite perspective
- **Who Gets What and Why**, Alvin Roth, matching markets (Nobel)

Closing Reflection

Thinking Strategically . . .

- "The supreme art of war is to subdue the enemy without fighting." - Sun Tzu, The Art of War
- "In the real world, the game isn't broken. The players are human." - Kahneman / Tversky, Prospect Theory Reflection
- Understand the incentives.
- Respect the players.
- May your equilibrium always be stable.